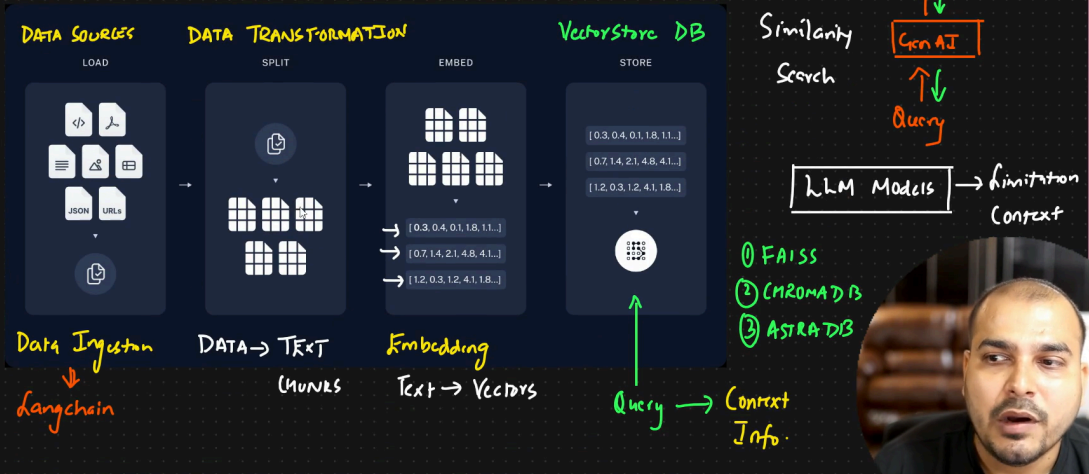


Important Components Of Langchain

① RAG ÷ Retrieval Augmented Generation



Here are different ways to load documents using langchain. URI changed, but you can research on this.

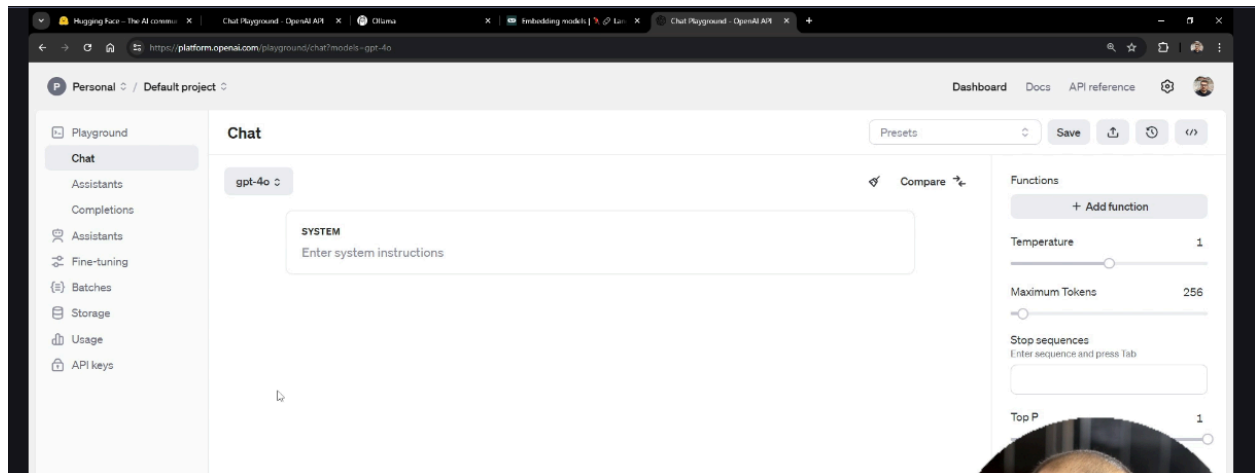
The screenshot shows the LangChain v0.2 documentation page for Document Loaders. The page has a dark theme with a sidebar on the left containing a list of document loaders: Document Processing, Org-mode, Pandas DataFrame, Pebble Safe DocumentLoader, Polars DataFrame, Psychic, PubMed, PySpark, Quip, ReadTheDocs Documentation, Recursive URL, Reddit, Roam, Rockset, rspace, RSS Feeds, RST, and a search bar. The main content area has a blue header 'Document loaders' and a section 'Features'. Below this, a table lists document loaders and their descriptions. The table has three columns: Document Loader, Description, and a partially visible 'Laz' column. The table lists AZLyricsLoader, AcreomLoader, AirbyteCDKLoader, and AirbyteGongLoader.

Document Loader	Description	Laz
AZLyricsLoader	Load AZLyrics webpages.	
AcreomLoader	Load acreom vault from a directory.	
AirbyteCDKLoader	Load with an Airbyte source connector implemented using the CDK.	
AirbyteGongLoader	Load from Gong using an Airbyte source connector.	

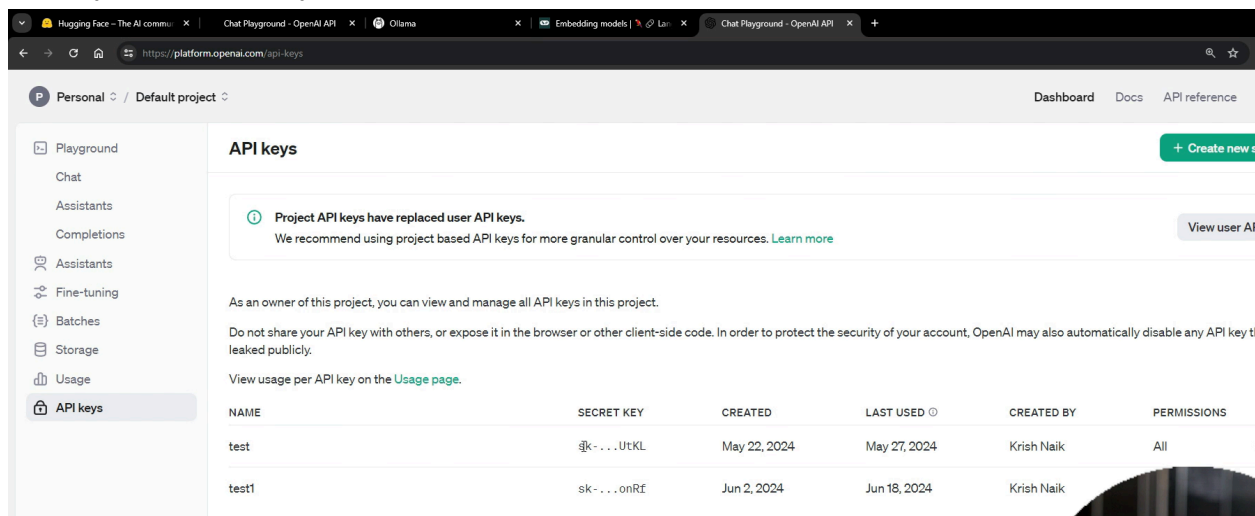
Embedding techniques

The image shows handwritten notes on a dark background. The word 'Embedding' is written in yellow. Below it, 'Text → Vectors' is written in white. To the right, 'Query → Context Info.' is written in yellow. At the bottom, there are three numbered items in green: '1) OpenAI → PAID', '2) OLLama', and '3) Huggingface → Open Source'. A green arrow points from '2) OLLama' to '1) OpenAI'.

For open ai, you have to invest 5 \$



create your own api key



Ollama embeddings(refer to the python code)

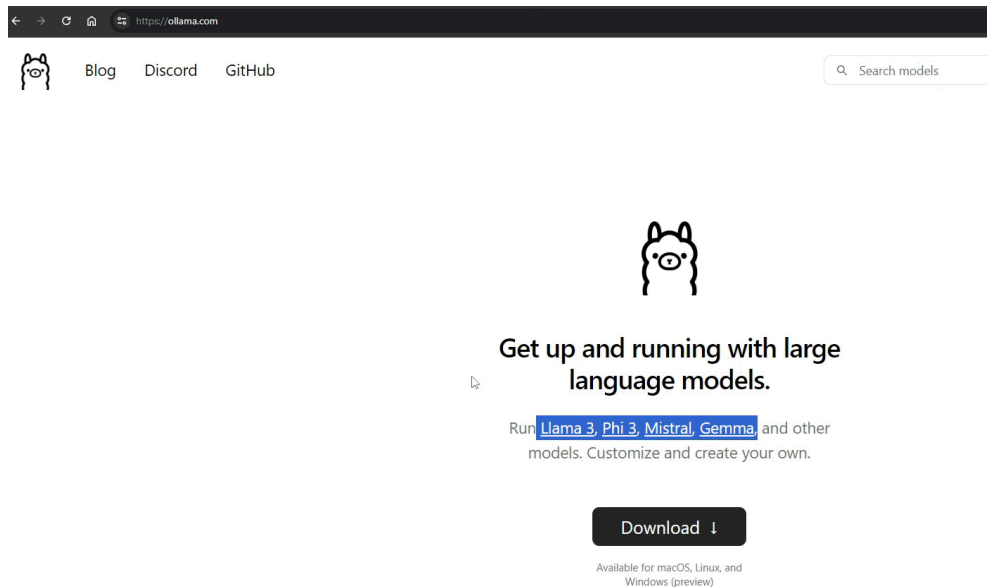
if you do not want to use openAI api, and you want to use open source models and use embedded techniques is with the platform called ollama

Ollama is an open-source tool that enables users to run large language models (LLMs) locally on their own computers without relying on cloud services or sending data to third parties. It stands for Omni-Layer Learning Language Acquisition Model and functions as a local AI orchestration platform that simplifies the deployment of pre-trained models like Llama 3, Mistral, and Phi-3 through a command-line interface (CLI) and an API.

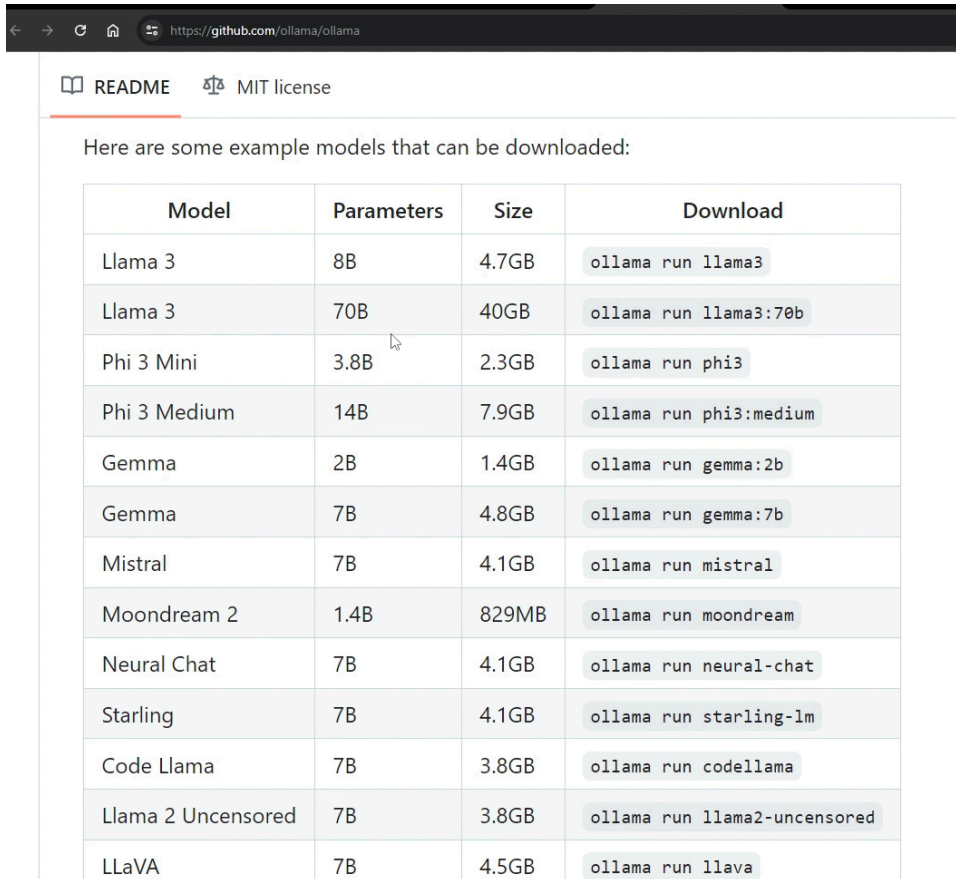
By running models directly on the user's device, Ollama ensures 100% local data privacy, allows for offline usage, and eliminates recurring subscription or per-token fees. It optimizes performance using quantization to reduce computational requirements, making it feasible to run powerful AI models on consumer-grade hardware.

- **Privacy:** Data never leaves the local device, unlike cloud-based AI where data is processed by external servers.
- **Cost:** It is free to use after purchasing the necessary hardware, avoiding ongoing API costs.
- **Ease of Use:** Users can pull and run models with simple commands (e.g., `ollama run llama3`) without extensive machine learning expertise.
- **Compatibility:** It supports macOS, Linux, and Windows (preview), and integrates with third-party GUIs like Open WebUI.
- **Use Cases:** Ideal for developers, researchers, and regulated industries (e.g., healthcare, finance) requiring secure, customizable, and offline AI solutions.

ollama is a platform where you can use open source LLM models.



if you go to github of this, it will support all open source models



after you download it, you can select the model from the list and run it. for example

```
Microsoft Windows [Version 10.0.19045.4529]
(c) Microsoft Corporation. All rights reserved.

C:\Users\win10>ollama run gemma:2b
>>> hi
Hello! 🌟 It's nice to hear from you. What can I do for you today? 🌟

>>> who are you
I am a large language model, trained by Google. I am capable of engaging in a wide range of conversations on
various topics. How may I assist you today?

>>> Send a message (/? for help)
```

you download another model also for example

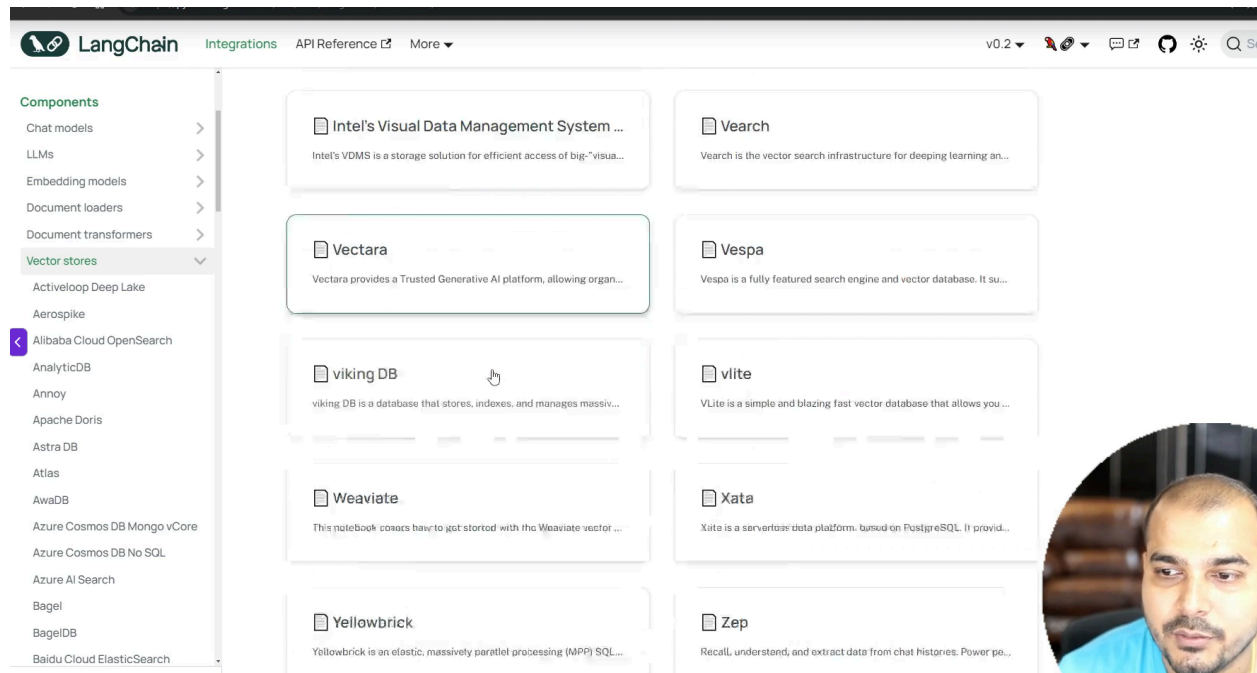
```
C:\Users\win10>ollama pull mxbai-embed-large  
pulling manifest  
pulling 819c2adf5ce6... 85% [██████████] 567 MB/669 MB 5.0 MB/s 20s
```

Hugging face embeddings

go to huggingface.co and create an account

Here create your access token for accessing the DB

you can also get a vectorDB of your choice from here



Here is the link for other DB's

Chroma

Chroma is a AI-native open-source vector database focused on developer productivity and happiness. Chroma is licensed under Apache 2.0.

<https://python.langchain.com/v0.2/docs/integrations/vectorstores/>

Hugging Face is a leading platform for pre-trained AI models and libraries, primarily known for its Transformers library, which simplifies accessing state-of-the-art models for tasks like text generation, translation, and sentiment analysis. It functions as a model store, offering a vast repository of open-source models, datasets, and tools for fine-tuning and deploying machine learning applications.

LangChain is a framework designed for building applications that rely on dynamic interactions with language models, emphasizing chaining together components like prompts, memory, and APIs to create complex workflows. It acts as a builder, enabling developers to integrate AI models (often from Hugging Face) into chatbots, agents, and multi-step reasoning systems with features like context retention and tool usage.

The two tools are highly complementary and often used together:

- Integration: A dedicated partner package, `langchain_huggingface`, was jointly launched to seamlessly integrate Hugging Face models into LangChain workflows.
- Workflow: Hugging Face provides the models and inference capabilities, while LangChain structures the application logic around them.
- Use Cases: Hugging Face is ideal for finding and fine-tuning models, whereas LangChain is preferred for creating interactive, multi-step applications like conversational agents or knowledge retrieval systems.